



Spring with Maven

1. What is **Maven**

Maven is a **build automation and dependency management tool** for Java projects.

It helps developers:

- Download required libraries (dependencies)
- Build the project
- Compile source code
- Run tests
- Package the application (JAR/WAR)

Maven is a build tool that automates project building, dependency management, testing, and packaging using a configuration file called `pom.xml`

Why do we need Maven?

Without Maven:

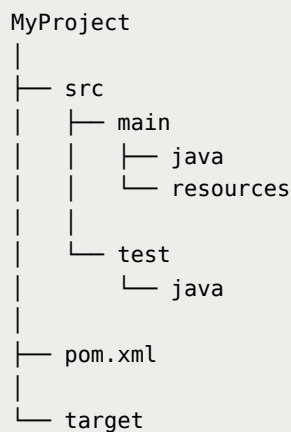
- Download JAR files manually.

- Copy them into the project.
- Update versions manually.

With Maven:

- Just specify dependencies in `pom.xml`.
- Maven downloads everything automatically.

Maven Project Structure



Folder Description

Folder/File	Purpose
<code>src/main/java</code>	Java source code
<code>src/main/resources</code>	Configuration files
<code>src/test/java</code>	Test classes
<code>pom.xml</code>	Maven configuration file
<code>target</code>	Compiled output (JAR/WAR files)

2. What is `pom.xml` ?

`pom.xml` stands for **Project Object Model**.

It contains:

- Project information

- Dependencies
- Plugins
- Build configuration

Example:

```
<project>

<groupId>com.example</groupId>

<artifactId>demo</artifactId>

<version>1.0</version>

</project>
```

Important Elements in `pom.xml`

1. groupId

Unique name of the organization.

```
<groupId>com.example</groupId>
```

2. artifactId

Project name.

```
<artifactId>spring-demo</artifactId>
```

3. version

Project version.

```
<version>1.0.0</version>
```

4. packaging

Output type.

```
<packaging>jar</packaging>
```

or

```
<packaging>war</packaging>
```